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Inflammatory Breast Cancer and Racial Disparities in a SEER Database Analysis of Survival Outcomes

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Supplementary Materials

Supplementary tables are labeled as Tables S1–S8 and
Supplementary figures are labeled as Figures S1–S4.

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Table S1: 3-Year Breast Cancer-Specific Survival (BCSS) Probabilities and Median Survival Times

Subgroup	N	Events	3-Year BCSS	95% CI	Median (Months)	Log-rank P
Overall	3,174	1,187	80.5%	(79.1–82.0)	Not Reached	–
IBC Status						<0.001
Non-IBC LABC	2,695	985	81.5%	(79.9–82.9)	Not Reached	
IBC	479	202	75.9%	(72.1–79.9)	Not Reached	
Race						<0.001
White	2,434	908	81.2%	(79.6–82.7)	Not Reached	
Black	363	160	70.9%	(66.4–75.7)	109	
API	377	119	85.6%	(82.3–89.0)	Not Reached	
Molecular Subtype						<0.001
HR+/HER2- (Luminal A)	1,882	676	85.1%	(83.5–86.8)	Not Reached	
HR+/HER2+ (Luminal B)	448	134	88.4%	(85.5–91.5)	Not Reached	
HR-/HER2+ (HER2 Enriched)	299	84	82.9%	(78.7–87.4)	Not Reached	
HR-/HER2- (TNBC)	545	293	56.5%	(52.4–60.9)	52	

Table S2: Sensitivity Analysis: IBC Effect on Mortality by Statistical Approach

Model Type	HR (95% CI)	p-value	Interpretation
Cause-Specific Cox (BCSS)[1]	1.415 (1.196–1.675)	<0.001	Primary analysis
Overall Survival (OS)[2]	1.315 (1.139 –1.519)	<0.001	Secondary outcome
Competing Risks (Fine-Gray)[3]	1.377 (1.163–1.631)	<0.001	Accounts for competing causes

[1]Cox proportional hazards model with BCSS as outcome; censoring at non-cancer death
[2]Cox proportional hazards model with overall mortality as outcome
[3]Fine-Gray subdistribution hazard model accounting for competing causes of death
All models adjusted for age, race, subtype, grade, stage, surgery, chemotherapy, radiation, and marital status

Table S3: Model Selection Criteria for Parametric Survival Models (AIC/BIC)

Distribution Model	AIC	BIC	AIC Rank
Log-Normal	14,699.36	14,729.67	1 (Best)
Exponential	14,812.75	14,837.00	2
Weibull	14,813.71	14,844.02	3

Table S4: Multivariable Cox Proportional Hazards Model Results: Race Comparisons by Molecular Subtype

Molecular Subtype / Race	HR	95% CI	*p*-value
Triple-Negative Breast Cancer (TNBC)			
Black vs. White	1.334	(0.977, 1.821)	0.0702
API vs. White	1.028	(0.717, 1.474)	0.8804
Luminal A (HR+/HER2−)			
Black vs. White	1.129	(0.876, 1.456)	0.3474
API vs. White	0.704	(0.539, 0.920)	0.0101
Luminal B (HR+/HER2+)			
Black vs. White	1.120	(0.629, 1.994)	0.7012
API vs. White	0.562	(0.290, 1.091)	0.0886
HER2-Enriched (HR−/HER2+)			
Black vs. White	1.693	(0.905, 3.168)	0.0995
API vs. White	1.185	(0.594, 2.361)	0.6301

Note: White women served as the reference group. Models adjusted for IBC status, age group, AJCC stage, marital status, surgery, chemotherapy, tumor grade, and radiation therapy, with stratification as needed. Estimates are from separate subtype-specific models; formal race-by-molecular-subtype interaction results are reported in Table S8.

Table S5: Comparison of Time Ratios (TR) and p-values Across Log-normal, Weibull, and Exponential Accelerated Failure Time (AFT) Models for Breast Cancer-Specific Survival

Covariate	Log-normal TR	p-value	Weibull TR	p-value	Exponential TR	p-value
IBC status						
IBC vs Non-IBC LABC	0.742 (0.626, 0.879)	<0.001	0.719 (0.613, 0.844)	<0.001	0.711 (0.601, 0.841)	<0.001
Race						
Black vs White	0.844 (0.703, 1.012)	0.068	0.813 (0.689, 0.960)	0.014	0.808 (0.678, 0.963)	0.017
API vs White	1.274 (1.059, 1.534)	0.010	1.213 (1.011, 1.455)	0.038	1.225 (1.011, 1.484)	0.038
Age group						
Post-menopausal vs Pre-menopausal	0.834 (0.723, 0.962)	0.013	0.817 (0.711, 0.939)	0.004	0.812 (0.702, 0.940)	0.005
Molecular Subtype						
HR+/HER2+ vs HR+/HER2-	1.306 (1.090, 1.564)	0.004	1.263 (1.054, 1.513)	0.012	1.277 (1.055, 1.545)	0.012
HR-/HER2+ vs HR+/HER2-	1.186 (0.953, 1.475)	0.126	1.303 (1.037, 1.636)	0.023	1.321 (1.039, 1.680)	0.023
HR-/HER2- vs HR+/HER2-	0.434 (0.371, 0.508)	<0.001	0.523 (0.452, 0.605)	<0.001	0.509 (0.437, 0.593)	<0.001
Tumor Grade						
Grade II vs Grade I	0.833 (0.658, 1.056)	0.131	0.831 (0.657, 1.052)	0.124	0.827 (0.645, 1.060)	0.133
Grade III/IV vs Grade I	0.561 (0.441, 0.714)	<0.001	0.606 (0.477, 0.770)	<0.001	0.595 (0.462, 0.765)	<0.001
AJCC Stage						
Stage IIIC vs Stage IIIB	0.727 (0.640, 0.826)	<0.001	0.698 (0.616, 0.790)	<0.001	0.687 (0.603, 0.784)	<0.001
Marital status						
Unmarried vs Married	0.872 (0.750, 1.013)	0.074	0.894 (0.776, 1.030)	0.120	0.889 (0.765, 1.032)	0.121
Unknown vs Married	0.729 (0.550, 0.967)	0.029	0.709 (0.552, 0.911)	0.007	0.702 (0.539, 0.915)	0.009
Primary surgery type						
Mastectomy vs Segmental	0.710 (0.596, 0.847)	<0.001	0.722 (0.604, 0.862)	<0.001	0.711 (0.590, 0.857)	<0.001
No Surgery vs Segmental	0.295 (0.230, 0.377)	<0.001	0.343 (0.272, 0.433)	<0.001	0.329 (0.259, 0.419)	<0.001
Chemotherapy						
Yes vs No	1.913 (1.641, 2.230)	<0.001	1.934 (1.678, 2.229)	<0.001	1.968 (1.696, 2.284)	<0.001
Radiation						
Yes vs No	1.482 (1.283, 1.711)	<0.001	1.362 (1.191, 1.557)	<0.001	1.378 (1.196, 1.587)	<0.001
Unknown vs No	1.322 (0.984, 1.776)	0.064	1.226 (0.929, 1.618)	0.149	1.236 (0.922, 1.656)	0.157

Table S6: Propensity-Score-Matched Black–White Comparisons by Inflammatory Breast Cancer Status

Cohort	Black	White	Total	BCSS deaths	Maximum |SMD|	Balance	HR (95% CI)	*p*-value
IBC	50	50	100	46	0.207	No	1.122 (0.635–1.984)	0.692
Non-IBC LABC	305	305	610	243	0.074	Yes	1.369 (1.063–1.764)	0.015

Note: Propensity-score matching used 1:1 greedy nearest-neighbor matching without replacement and a standardized caliper width of $0.20\,\mathrm{SD}(\widehat{e})$, where $\widehat{e}$ denotes the estimated propensity score probability. Acceptable balance required all absolute standardized mean differences (SMDs) to be below 0.10. Hazard ratios (HRs) compare Black with White women using Cox proportional hazards models with robust standard errors.

Table S7: Statistical Precision of the Black–White Comparison Within the Inflammatory Breast Cancer Cohort

Measure	Value
Total Black and White IBC patients	433
Black IBC patients	54
White IBC patients	379
Breast cancer-specific deaths	188
Proportion of patients who were Black	12.47%
Detectable HR above 1 with 80% power	1.856
Detectable HR below 1 with 80% power	0.539
Observed non-IBC Black vs. White HR	1.281
Approximate power to detect HR 1.281 in the IBC subgroup	20.17%

Note: Calculations assumed a two-sided significance level of 0.05. The HR of 1.281 was the adjusted Black–White estimate in the non-IBC LABC cohort and was used as a benchmark. The detectable HR of 0.539 is the reciprocal of 1.856.

Table S8: Formal Race-by-Molecular-Subtype Interaction Analysis for Breast Cancer-Specific Survival

Interaction term	Ratio of HRs	95% CI	*p*-value
Black-by-molecular-subtype interactions			
Black-by-Luminal B	0.920	(0.510–1.659)	0.7810
Black-by-HER2-enriched	1.568	(0.814–3.019)	0.1784
Black-by-triple-negative breast cancer	1.138	(0.770–1.683)	0.5167
API-by-molecular-subtype interactions			
API-by-Luminal B	0.856	(0.423–1.731)	0.6645
API-by-HER2-enriched	1.435	(0.695–2.961)	0.3285
API-by-triple-negative breast cancer	1.496	(0.960–2.329)	0.0750

Note: Estimates came from a common Cox model with White women and Luminal A as reference categories. The model adjusted for age group, AJCC stage, marital status, and chemotherapy, with stratification by IBC status, grade, surgery, and radiation. The omnibus interaction was nonsignificant by the Wald test ($\chi^2(6) = 6.145,\ p = 0.407$) and likelihood-ratio test ($\chi^2(6) = 5.998,\ p = 0.423$).

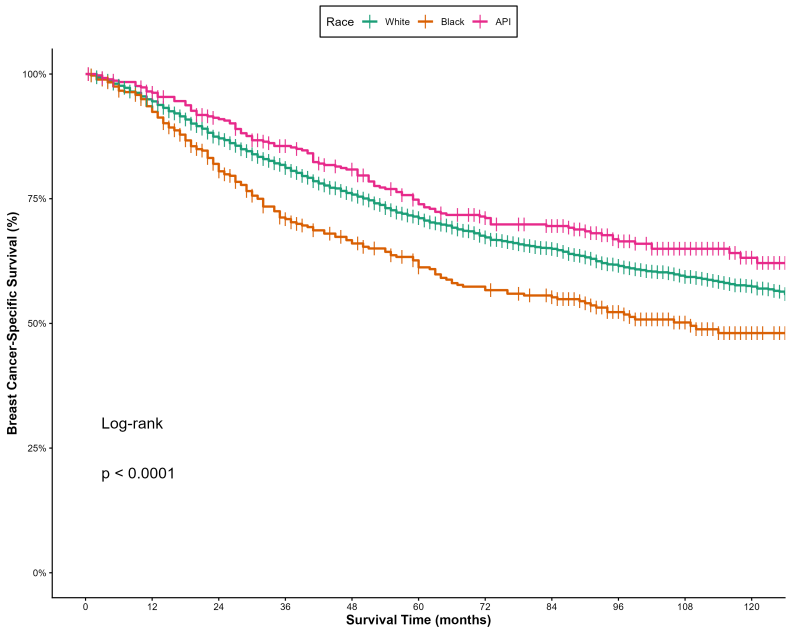


Fig. S1: Breast Cancer-Specific Survival by Race

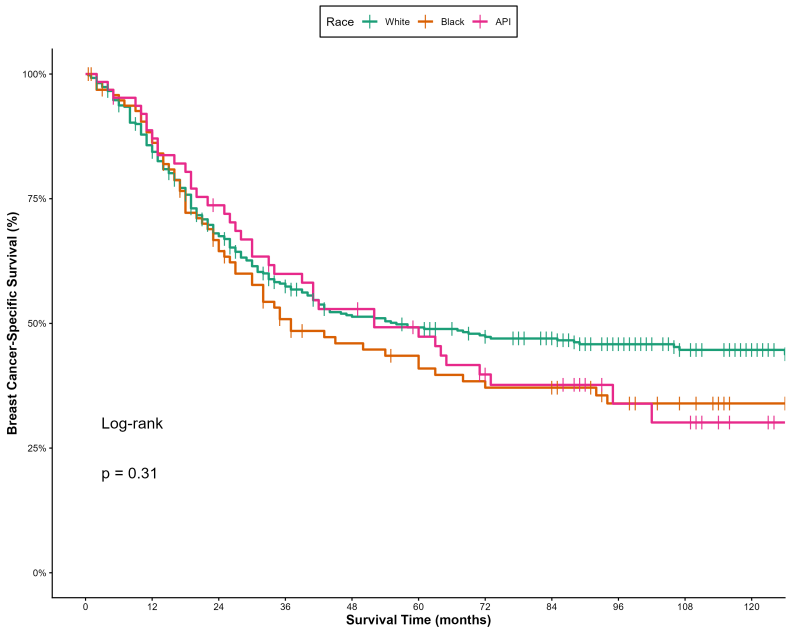


Fig. S2: Survival by Race in the Triple-Negative (TNBC) Subgroup

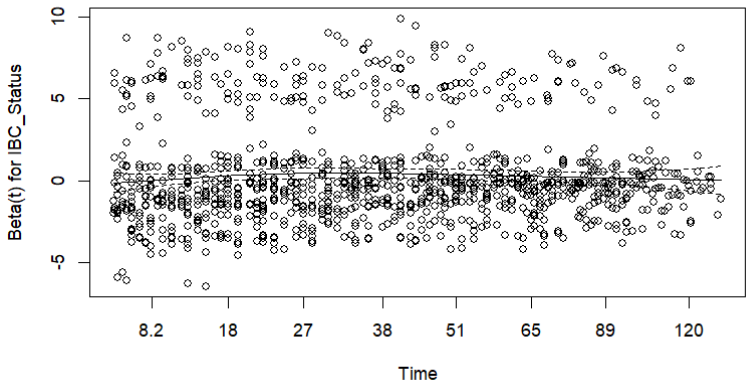
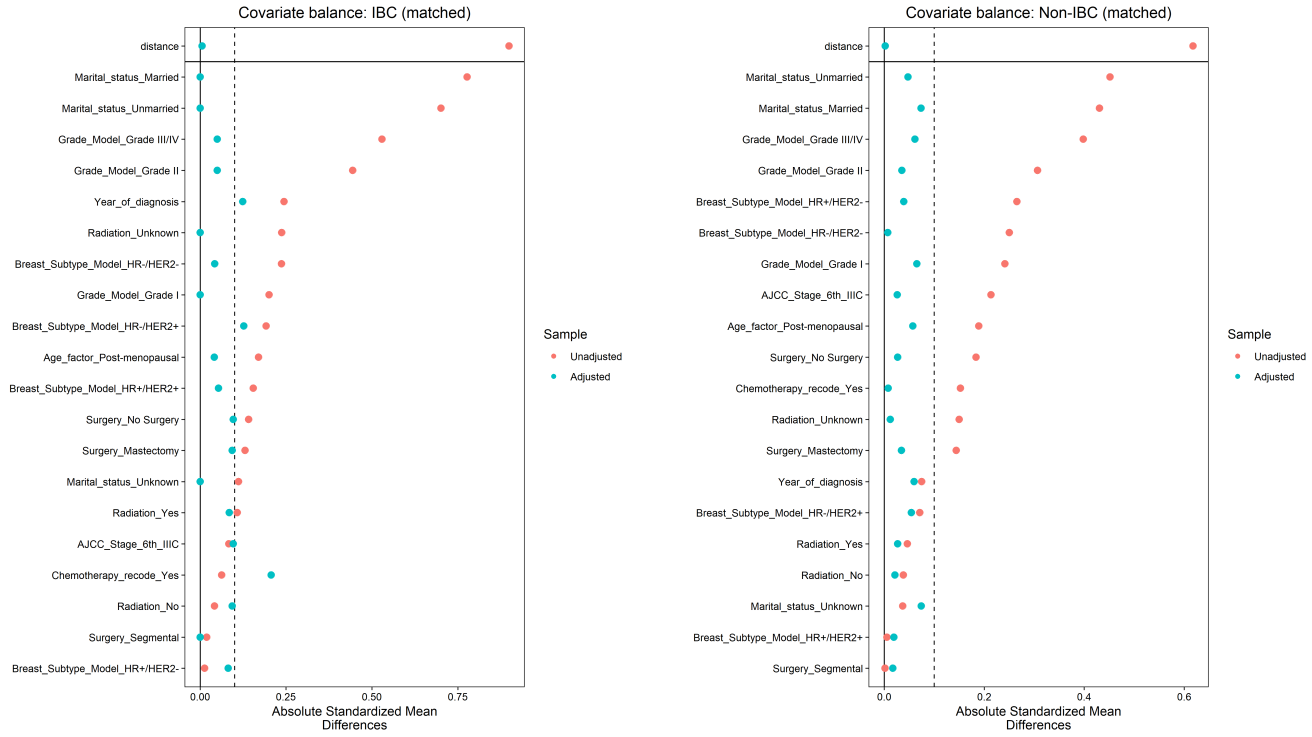


Fig. S3: Scaled Schoenfeld residuals for IBC status in the stratified Cox model.

Note. In the initial multivariable Cox model, the overall tumor grade term showed evidence of nonproportionality ($\chi^2 = 93.095$, $df = 2$, $p < 0.001$). Tumor grade was therefore included as a stratification factor in the sensitivity analysis, allowing separate baseline hazards across grade categories and avoiding a common proportional hazards constraint for grade. In the stratified model, the global Schoenfeld test for the remaining regression covariates was nonsignificant ($\chi^2 = 3.958$, $df = 8$, $p = 0.861$), and the IBC status test was also nonsignificant ($\chi^2 = 0.025$, $df = 1$, $p = 0.874$).



(a) IBC Cohort

(b) Non-IBC LABC Cohort

Fig. S4: Covariate Balance for Propensity Score Matching